

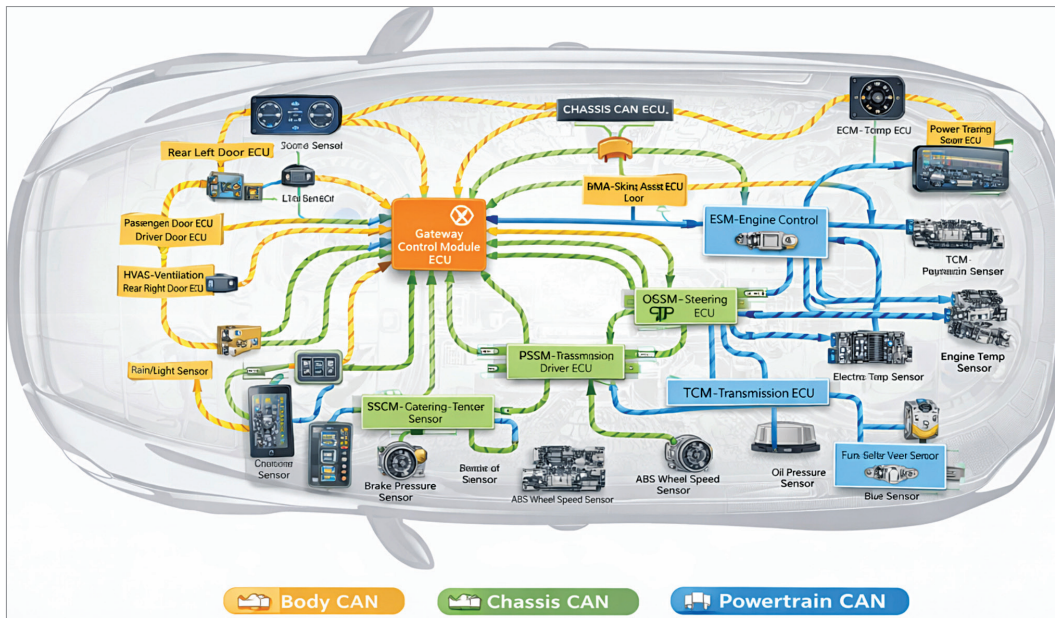


Fig. 2: Vehicle CAN network with interconnected ECUs



Fig. 3: Vehicle movement cases for doppler speed measurement

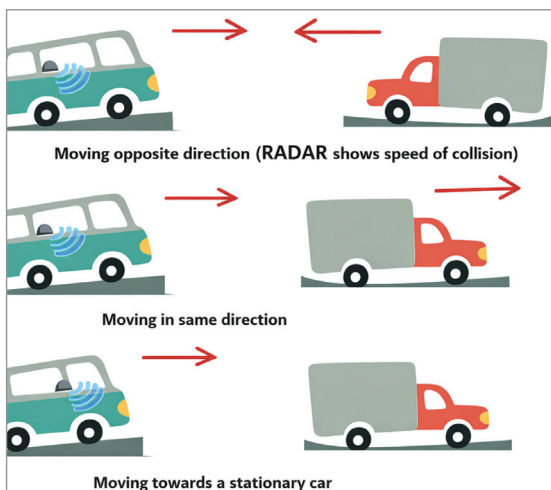


Fig. 4: Prototype testing setup showing the roadside environment and moving vehicles

Furthermore, the module features built-in IoT capabilities via Wi-Fi, enabling wireless monitoring of all radar and IMU parameters

As shown in Fig. 1, a CAN bus radar is mounted on the vehicle's bonnet. On the right side, the radar data is displayed wirelessly on the IoT dashboard

provided by the module itself. The components required to build this system are listed in the Bill of Materials table.

CAN bus design

Unlike traditional communication systems that require separate wires for each device, the CAN bus connects all ECUs and sensors using only two twisted wires: CAN High and CAN Low. In a multi-

master configuration, all ECUs share the same bus. Any ECU can transmit messages at any time, and every node on the bus receives the data. Each ECU then decides whether to process or ignore the message based on its identifier.

The twisted-pair wiring effectively rejects electromagnetic interference, ensuring reliable communication in noisy vehicle environments. Data is transmitted by varying the voltage difference between the two wires: a dominant state (logic 0) produces a 2V differential, while the recessive state (logic 1) keeps both wires at approximately 2.5V. This simple and robust design enables efficient, real-time data sharing among all devices with minimal wiring.

How doppler radar works

Doppler radar works on the Doppler effect, which is the change in a wave's frequency due to relative motion between the source and the observer. In a vehicle radar system, the radar transmits microwave signals towards moving or stationary objects. When these signals are reflected back, any relative motion causes a shift in the returned frequency. If an object is approaching the vehicle, the reflected frequency increases (positive shift), while if